BRAHMA TEJA PUNNAMARAJU

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PROFESSIONAL SUMMARY

- *7 years in data science, analytics, and machine learning, including 2+ years building Generative AI applications (LLMs, RAG, SLMs), across consulting, nonprofit, policy, and automotive sectors.*
- *Deployed GenAI solutions with Snowflake Cortex, LangChain, Azure OpenAI (GPT-3.5 Turbo), Llama, and SLMs (Phi-4, Gemma, Mistral), achieving up to 94% response accuracy and cutting manual document review by ~60%.*
- *Built scalable data and ML pipelines and optimization models using Python, SQL, Snowflake Snowpark, and Azure, delivering ~$10M in projected annual savings and consolidating 110 legacy reports into 9 Power BI dashboards.*
- *Built predictive, segmentation, and forecasting models (attrition, latent class, regression, time series) with scikit-learn and Azure ML, lifting donor retention 15%, CSAT from 3.5 to 4.2, and informing a product launch worth $120M in revenue.*
- *Strong foundation in statistical modeling, hypothesis testing, A/B testing, and optimization, with a track record of turning model outputs into business decisions for technical and non-technical stakeholders.*

PROFESSIONAL EXPERIENCE

KAVI GLOBAL **Illinois, USA**

Data Scientist *Sept 2024 - Present*

- **Data Pipelines & Optimization:** Built ETL pipelines and a simulation-based optimization model to right-size phlebotomist staffing across demand scenarios, projecting ~$10M in annual savings.
- **Dynamic Assignment Model:** Built a real-time assignment model matching phlebotomist availability to demand, improving coverage consistency by ~30% and reducing scheduling conflicts.
- **LLM Chatbot Optimization:** Raised Snowflake Cortex Analyst/Search chatbot accuracy to 94% through prompt engineering and refining text-to-SQL semantic model (YAML) configurations on Llama.
- **RAG Chatbot Prototype:** Built a scalable RAG chatbot (embeddings, vector search) for the Evangelical Lutheran Church in America over structured and unstructured data, achieving 85%+ retrieval accuracy in pilot testing.
- **Document Ingestion & Processing:** Built Snowpark (Python) ingestion workflows for large PDF and text corpora, cutting data retrieval latency by ~40% for downstream RAG search.
- **Semantic Search & Interface Design:** Implemented LangChain semantic search and a Streamlit chatbot UI, increasing user engagement by 25% across internal stakeholder test groups.
- **Retrieval & Response Tuning:** Tuned retrieval and semantic configurations across Cortex Analyst, Cortex Search, and Llama via iterative LLM evaluation, achieving a 92% match rate between user intent and generated responses.
- **SLM-Based Information Extraction:** Built an SLM pipeline (Phi-4, Gemma, Mistral) for a global electronics client to extract structured data from engineering drawings, cutting manual review effort by ~60%.
- **AI-Assisted Prototyping:** Built interactive prototypes with Google Antigravity during requirements gathering for CNH, validating needs with stakeholders upfront and cutting requirements cycles by ~40%.
- **BI Consolidation:** Consolidated 110 Qlik/Cognos reports into 9 Power BI dashboards for sales, partner programs, compliance, and other teams, shrinking the report footprint by ~92%.

COMMUNITY DREAMS FOUNDATION CORP **Illinois, USA**

Data Analyst (Remote) *Apr 2024 - Sept 2024*

- **Data Pipelines & KPI Monitoring:** Built Azure Data Factory pipelines feeding survey and social media data into a real-time Power BI KPI dashboard, contributing to a 30% improvement in learning outcomes.
- **Donor Attrition Prediction:** Built a donor attrition (churn) classification model in Azure ML using Synapse-based behavioral segmentation, enabling targeted outreach that increased donor retention by 15%.

EVIDENCE-TO-IMPACT COLLABORATIVE **Pennsylvania, USA**

Program Manager Intern *Sep 2023 - Mar 2024*

- **Data Profiling & Quality:** Profiled and mapped policy datasets in Azure SQL Database, standardizing formats for cost-benefit models—improving data accuracy by 27% and saving ~100 hours of data cleaning.
- **Dashboard Automation:** Designed SQL data models and automated Tableau dashboards, cutting manual reporting by 30% (~25 hours/month) and saving USD 50,000 annually.
- **LLM Application Deployment:** Built and deployed a GPT-3.5 Turbo–based chatbot on Azure OpenAI Service to answer policy research questions, reducing policy search time by 60%.

HYUNDAI MOTORS **Telangana, India**

Senior Business Analyst *Aug 2018 - June 2022*

- **Data-Driven Feature Discovery:** Analyzed 20 years of road accident data with Azure SQL and Power BI, identifying 12 high-risk scenarios that shaped autonomous vehicle feature requirements.
- **Customer Segmentation:** Built a customer segmentation model using latent class analysis to target customers with tailored products, raising CSAT from 3.5 to 4.2 and increasing repeat customers by 25%.
- **Product Analytics:** Led data-driven product comparisons using market research data to address customer concerns, lifting conversion rate from 15% to 18% and contributing ~USD 500,000 in incremental annual revenue.
- **Reporting Automation:** Automated the monthly car sales report across 600,000+ records using Advanced Excel and Power BI, cutting person-hours by 65% (~300 hours saved annually).
- **Demand Analysis & Product Launch:** Identified micro-SUV demand via SQL and Power BI analysis of customer data and pitched the product proposal, driving a launch with 100K bookings and $120M revenue in five months.
- **Sales & Market Forecasting:** Built statistical and time series forecasting models for market trends and sales, informing strategic planning and driving a 20% gain in sales efficiency and a 5% rise in market share.

CERTIFICATIONS

- PCEP – Certified Entry-Level Python Programmer certification *[Certification ID: kXgp.QSAC.E7uR]*
- PCAP –Associate Python Programmer certification [Certification ID: CFBh.s6KG.hRRV]

EDUCATION

Pennsylvania State University **State College, PA, USA**

Master of Business Analytics (STEM) *Aug 2023*

Koneru Lakshmaiah University **Vijayawada, India**

Bachelor of Technology in Mechanical Engineering *Jun 2018*

SKILLS AND EXTRACURRICULARS

- **Programming, ML & Statistics:** Python (Pandas, NumPy, scikit-learn, TensorFlow, Matplotlib), SQL, R, Jupyter, Machine Learning, Predictive Modeling, Feature Engineering, Classification, Regression, Clustering (k-means), Time Series Forecasting, Statistical Modeling, Hypothesis & A/B Testing, Recommendation Systems, Optimization (Gurobi), Simulation
- **Generative AI & NLP:** LLMs (GPT-3.5 Turbo, Llama), SLMs (Phi-4, Gemma, Mistral), RAG, Prompt Engineering, Embeddings, Semantic & Vector Search, Text-to-SQL, NLP, Information Extraction, LLM Evaluation, LangChain, Snowflake Cortex (Analyst, Search), Azure OpenAI Service, Streamlit, REST APIs, AI-Assisted Prototyping (Google Antigravity)
- **Data Engineering, Cloud & BI:** ETL/ELT Pipelines, Data Modeling, Model Deployment, Snowflake (Snowpark), Azure (Data Factory, Synapse Analytics, Machine Learning, Databricks, Data Lake, SQL Database), Redshift, Dagster, Power BI, Tableau, Qlik, Cognos, Advanced Excel, Web Scraping (BeautifulSoup)

ACADEMIC PROJECTS

- **Artist based recommendation system (Spotify dataset):** Built an artist-based recommendation engine using EDA and k-means clustering on 170K+ Spotify tracks, grouping artists into 8 listener-similarity clusters.
- Modeled state-level drivers of diabetes and obesity using regression and time series forecasting, producing recommendations projected to save health insurers USD 264M per year.